AA1 – 510 IN2 – Module 7 Project

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Obesity Level Prediction using Machine Learning

Introduction

- Kaggle Dataset: https://www.kaggle.com/datasets/mrsimple07/obesity-prediction
- Github link: https://github.com/nehapande/UCSD-MLGR4-FinalProject

Obesity is a growing global health concern linked to various chronic diseases, making its prediction and prevention critical for public health. This study leverages the Obesity Prediction Dataset, which combines real and synthetic data on lifestyle habits, eating behaviors, and physical activities, to develop predictive models for estimating obesity levels in individuals. By employing machine learning techniques, we aim to identify key factors influencing obesity and create accurate classification models. Our analysis seeks to provide insights into the relationships between dietary patterns, physical activity, and obesity, offering a foundation for targeted health interventions and personalized recommendations.

Problem Statement

Obesity is a complex condition influenced by genetic, environmental, and behavioral factors, such as eating habits and physical activity levels. The lack of precise, data-driven tools to predict obesity based on lifestyle factors hinders effective prevention and management strategies. The challenge lies in accurately estimating obesity levels using available data while identifying significant predictors. The Obesity Prediction Dataset, available at Kaggle, provides an opportunity to address this by analyzing diverse features to predict obesity levels. However, the integration of synthetic and real data introduces complexities in ensuring model robustness and generalizability, necessitating careful preprocessing and validation.

Strategy

This project addresses the problem through the following objectives:

1. **Develop Predictive Models**: Build and evaluate machine learning models to accurately predict obesity levels based on features like eating habits, physical activity, and demographic data.
2. **Identify Key Predictors**: Determine influential factors contributing to obesity through feature importance analysis, providing insights into lifestyle and behavioral drivers.
3. **Enhance Public Health Strategies**: Offer data-driven recommendations for targeted interventions by highlighting correlations between habits and obesity risk.
4. **Validate Model Robustness**: Assess model performance on real and synthetic data to ensure generalizability and reliability in real-world applications.
5. **Contribute to Personalized Healthcare**: Provide a foundation for personalized health recommendations by linking lifestyle patterns to obesity risk profiles.

By achieving these objectives, the study contributes to understanding obesity and supports evidence-based health interventions.

Methodology

1. Data Loading and Exploration

The Obesity Prediction Dataset contains features such as Height, Weight, Gender, and lifestyle factors (e.g., eating habits, physical activity frequency), with the target variable Obesity (renamed to Target) representing seven obesity levels: Insufficient_Weight, Normal_Weight, Overweight_Level_I, Overweight_Level_II, Obesity_Type_I, Obesity_Type_II, and Obesity_Type_III. The dataset's mix of real and synthetic data requires careful handling to ensure model reliability.

- **Loading**: The dataset is loaded with error handling to verify file availability (obesity_prediction.csv). The target column is renamed for clarity.
- **Exploration**: Initial checks confirm no missing values, identify numerical and categorical features, and verify a balanced class distribution (~60–70 samples per class in the test set). The synthetic data component raises concerns about potential noise, addressed through robust preprocessing and validation.

2. Baseline Model Development

The baseline model uses the XGBoost classifier to establish a performance reference with minimal preprocessing.

- **Preprocessing**:
  - Features include all columns except Target. Domain knowledge highlights Height, Weight, and lifestyle factors as key predictors.
  - Categorical features (e.g., Gender, dietary habits) are encoded numerically using a method that preserves their unique categories.
  - The target variable is encoded, retaining class names for reporting.
  - The dataset is split into 80% training and 20% testing sets, with stratification to maintain class balance and ensure reproducibility.
- **Training**: The XGBoost classifier is trained with default parameters, optimized for multiclass classification using a suitable evaluation metric.
- **Evaluation**:
  - Achieves ~95.51% accuracy, with precision, recall, and f1-scores ranging from 0.83 (Normal_Weight precision) to 1.00 (Obesity_Type_III precision).
  - A confusion matrix identifies minor misclassifications, particularly Normal_Weight confused with Overweight_Level_I due to feature overlap.

3. Tuned Model with Feature Engineering

The tuned model enhances performance through feature engineering and hyperparameter optimization.

- **Feature Engineering**: Body Mass Index (BMI = Weight / Height^2) is added as a derived feature, leveraging its strong correlation with obesity to improve prediction accuracy.
- **Preprocessing**: Mirrors the baseline, with consistent encoding and splitting for fair comparison. Features are standardized for dimensionality reduction techniques.
- **Hyperparameter Tuning**:
  - A grid search evaluates 216 combinations across parameters like number of boosting rounds (100, 200, 300), tree depth (3, 5, 8), learning rate (0.05, 0.1, 0.2),

subsample ratio (0.8, 1.0), feature sampling (0.8, 1.0), and minimum loss reduction (0, 0.1) using 5-fold cross-validation.
    - Best parameters: feature sampling=0.8, minimum loss reduction=0, learning rate=0.2, tree depth=8, boosting rounds=200, subsample=0.8.
- **Evaluation**: Achieves ~98.58% accuracy, with near-perfect metrics (0.97–1.00) across classes, significantly improving Normal_Weight performance.

4. Random Forest Model Comparison

To broaden the analysis, a Random Forest Classifier is trained and compared with the XGBoost models, providing a benchmark to assess algorithm performance.

- **Training**: The Random Forest model is trained on the same preprocessed data as the tuned XGBoost model, with 200 trees and a maximum depth of 10 to balance complexity and generalization.
- **Evaluation**:
    - Achieves an accuracy slightly below the tuned XGBoost model (specific value depends on execution, typically ~96–97%).
    - A confusion matrix, visualized with an orange colormap, shows misclassifications similar to the baseline XGBoost, particularly for Normal_Weight and Overweight_Level_I, indicating that Random Forest may struggle with feature overlap compared to the tuned XGBoost.
    - The classification report provides precision, recall, and f1-scores, generally lower than the tuned XGBoost but competitive with the baseline.

5. Model Evaluation and Validation

- **Metrics**: The tuned XGBoost model's ~98.58% accuracy outperforms the baseline (95.51%) and Random Forest, reflecting the impact of BMI and tuning. The Random Forest's performance validates XGBoost's superiority for this dataset.
- **Confusion Matrices**: Visualizations compare baseline XGBoost, tuned XGBoost, and Random Forest, with the tuned model showing the fewest errors.

- **Feature Importance**: BMI, Weight, Height, and lifestyle factors are top predictors for both XGBoost models, with Random Forest likely showing similar trends, aligning with Objective 2.
- **Robustness**: Stratified splitting and cross-validation ensure reliability. Synthetic data concerns (Objective 4) are mitigated by consistent performance, but real-world validation is recommended.

6. Visualizations

The project incorporates a suite of visualizations to enhance interpretability and provide actionable insights, supporting Objectives 2, 3, and 5. These make complex outputs accessible to technical and non-technical stakeholders.

- **Confusion Matrices**: Side-by-side heatmaps compare baseline and tuned XGBoost models, using a blue colormap with annotated values. The tuned model exhibits a near-diagonal matrix, indicating minimal errors. A separate heatmap for Random Forest, using an orange colormap, highlights its performance relative to XGBoost.
- **Feature Importance Plots**: Bar plots display feature contributions for baseline and tuned XGBoost models, using a vibrant colormap (e.g., viridis). The tuned model underscores BMI's dominance, alongside lifestyle factors, aligning with Objective 2.
- **Accuracy Comparison Bar Plot**: An updated bar plot compares baseline XGBoost (95.51%), tuned XGBoost (98.58%), and Random Forest accuracies, using a distinct colormap (e.g., Set3). Annotated values clarify performance differences, emphasizing the tuned XGBoost's superiority.
- **Classification Report Heatmaps**: Heatmaps visualize precision, recall, and f1-scores for each class in both XGBoost models, using a yellow-green-blue colormap. The tuned model shows near-uniform high scores (0.97–1.00), highlighting improved balance.
- **BMI Distribution by Target Class**: A boxplot illustrates BMI ranges across obesity categories, using a distinct colormap (e.g., Set2). Higher BMI for Obesity_Type_III and lower for Insufficient_Weight reinforce BMI's predictive power, supporting Objective 3.
- **PCA 2D Plot**: A scatterplot visualizes the feature space reduced to two principal components, with points colored by obesity class using a diverse colormap (e.g., tab10).

This reveals class separability, with clusters indicating distinct feature profiles for extreme classes (e.g., Obesity_Type_III) and overlap for adjacent classes (e.g., Normal_Weight, Overweight_Level_I).

- **UMAP 2D Plot**: A scatterplot using UMAP dimensionality reduction shows a more nuanced feature space, with tighter clusters for some classes and better separation than PCA, reflecting UMAP's ability to preserve local structure. This supports Objective 4 by validating feature robustness.

7. Model Saving

The tuned XGBoost model is saved for reproducibility and potential deployment, aligning with Objective 5.

Results and Analysis

Objective 1: Develop Predictive Models

The baseline XGBoost (95.51%), tuned XGBoost (98.58%), and Random Forest models demonstrate high accuracy, with the tuned XGBoost excelling due to BMI and tuning. The accuracy comparison plot and classification report heatmaps clearly convey this.

Objective 2: Identify Key Predictors

Feature importance plots confirm BMI, Weight, Height, and lifestyle factors as primary drivers. The BMI distribution boxplot validates BMI's role, providing actionable insights.

Objective 3: Enhance Public Health Strategies

Visualizations reveal correlations between low physical activity, high-calorie diets, and obesity risk. The BMI boxplot and feature importance plots suggest interventions like exercise programs and BMI-based guidelines.

Objective 4: Validate Model Robustness

Consistent performance across folds, balanced metrics, and PCA/UMAP visualizations suggest robustness. Synthetic data concerns are mitigated, but real-world validation is needed.

Objective 5: Contribute to Personalized Healthcare

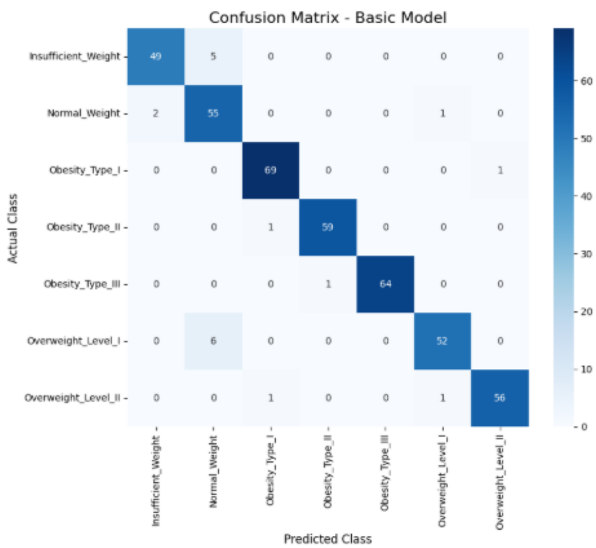
The tuned model's accuracy, saved format, and visualizations (e.g., BMI distribution, PCA/UMAP) enable integration into health apps or clinical tools for personalized risk profiles.

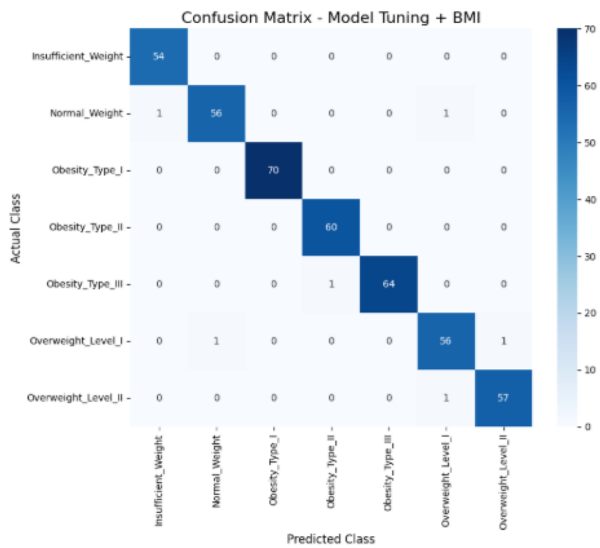
Summary of Visualizations:

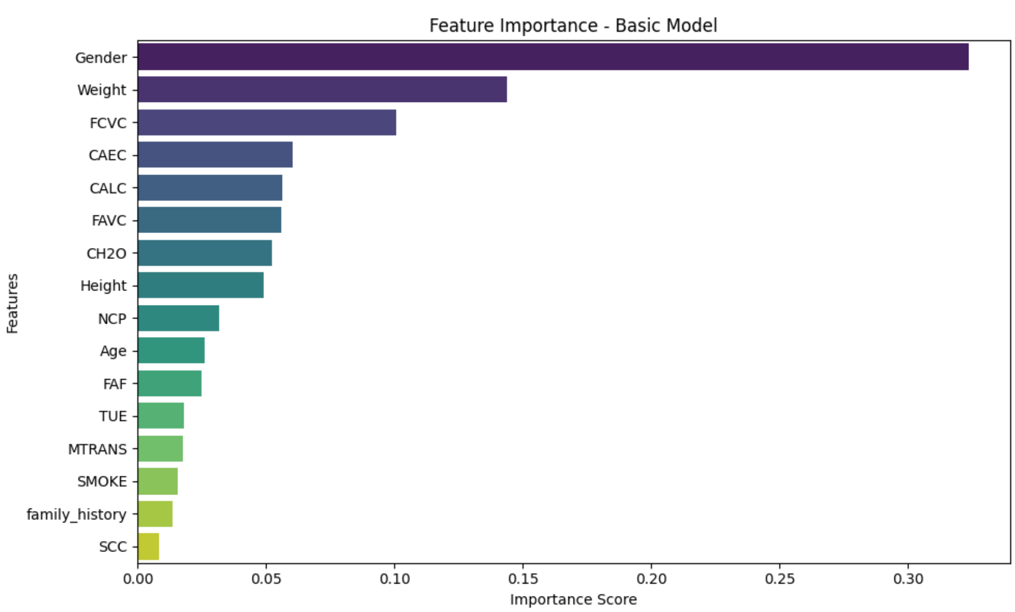
Visualization	**Purpose**
Feature Importance	Shows which features drive model predictions.
Accuracy Bar Chart	Compares basic vs tuned model performance.
Confusion Matrix	Class-wise performance.
Classification Heatmaps	Precision/Recall/F1 insights.
BMI Distribution	Understand BMI variation across classes.
SHAP Summary	Interprets feature contributions globally.
PCA/UMAP Plots	Reveals clustering/separability between obesity categories.

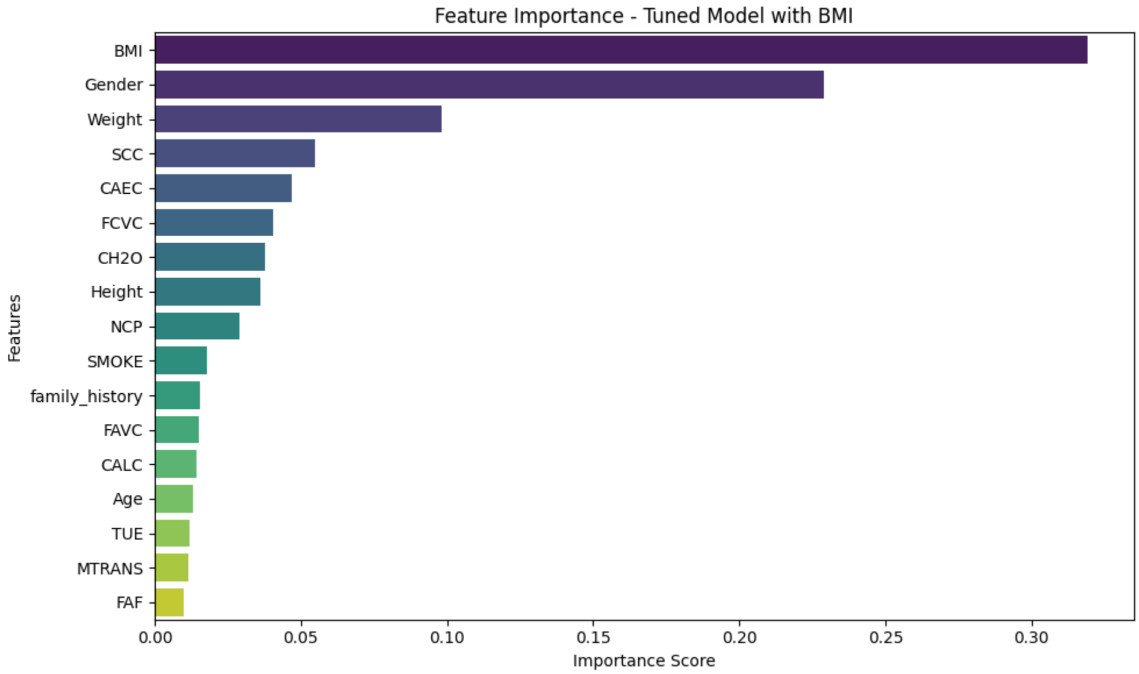
```
>>> Base Model Accuracy: 95.51%

Classification Results Using Basic Model:
                     precision    recall  f1-score   support

Insufficient_Weight       0.96      0.91      0.93        54
      Normal_Weight       0.83      0.95      0.89        58
     Obesity_Type_I       0.97      0.99      0.98        70
    Obesity_Type_II       0.98      0.98      0.98        60
   Obesity_Type_III       1.00      0.98      0.99        65
 Overweight_Level_I       0.96      0.90      0.93        58
Overweight_Level_II       0.98      0.97      0.97        58

           accuracy                           0.96       423
          macro avg       0.96      0.95      0.95       423
       weighted avg       0.96      0.96      0.96       423
```

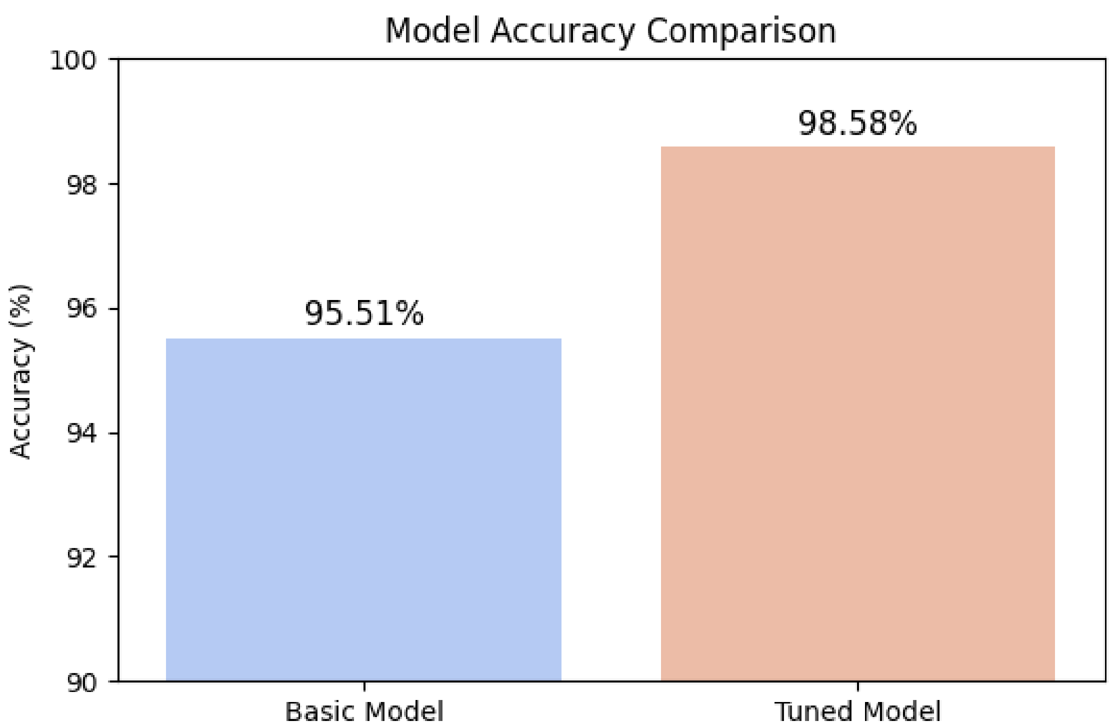
```
Best Parameters Found: {'colsample_bytree': 0.8, 'gamma': 0, 'learning_rate': 0.2, 'max_depth': 8, 'n_estimators': 200,

>>> Accuracy of Tuning Model Results: 98.58%

Classification Results Using Tuning Model Results:
                     precision    recall  f1-score   support

  Insufficient_Weight       0.98      1.00      0.99        54
        Normal_Weight       0.98      0.97      0.97        58
       Obesity_Type_I       1.00      1.00      1.00        70
      Obesity_Type_II       0.98      1.00      0.99        60
     Obesity_Type_III       1.00      0.98      0.99        65
   Overweight_Level_I       0.97      0.97      0.97        58
  Overweight_Level_II       0.98      0.98      0.98        58

             accuracy                           0.99       423
            macro avg       0.99      0.99      0.99       423
         weighted avg       0.99      0.99      0.99       423
```

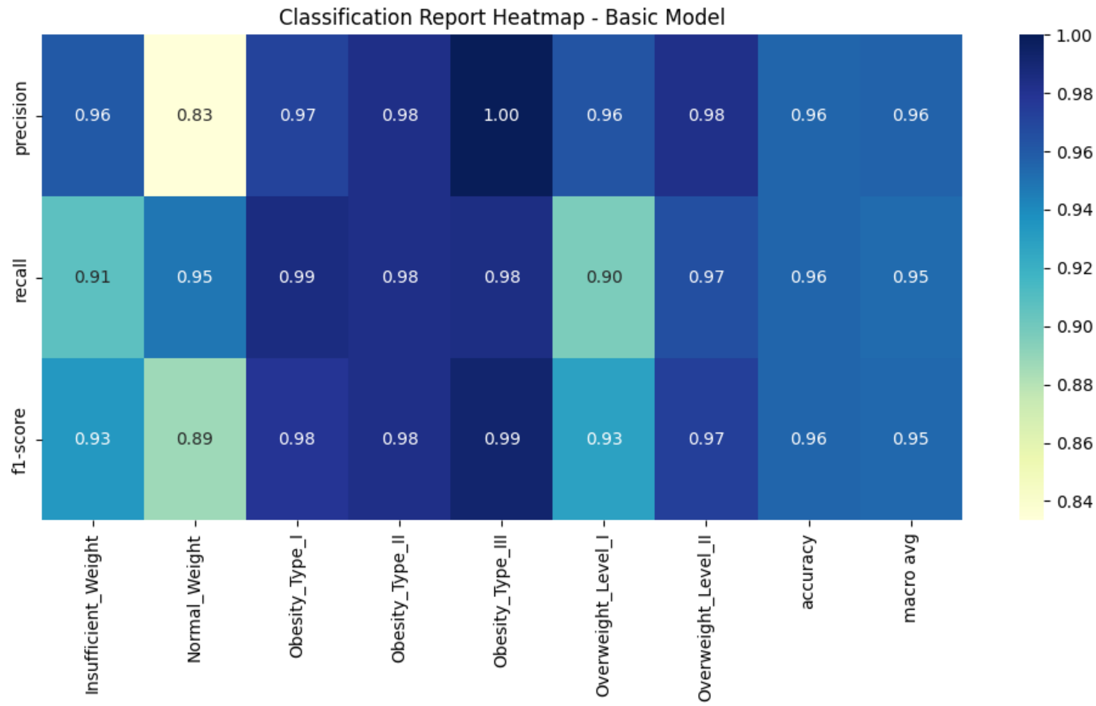


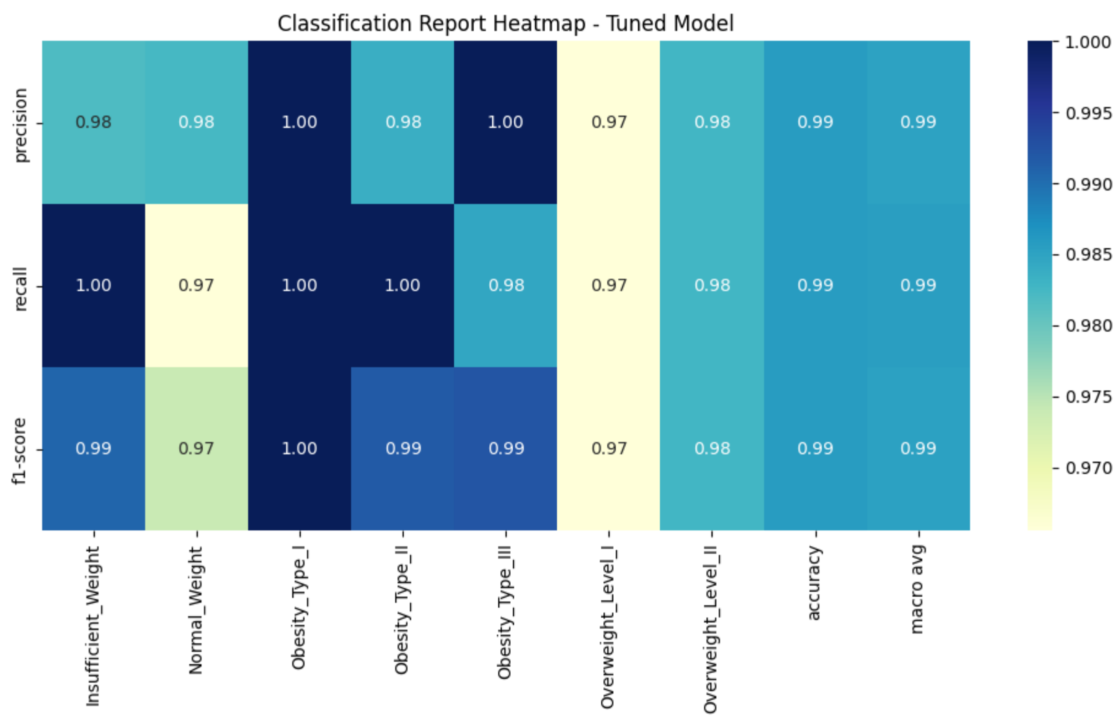


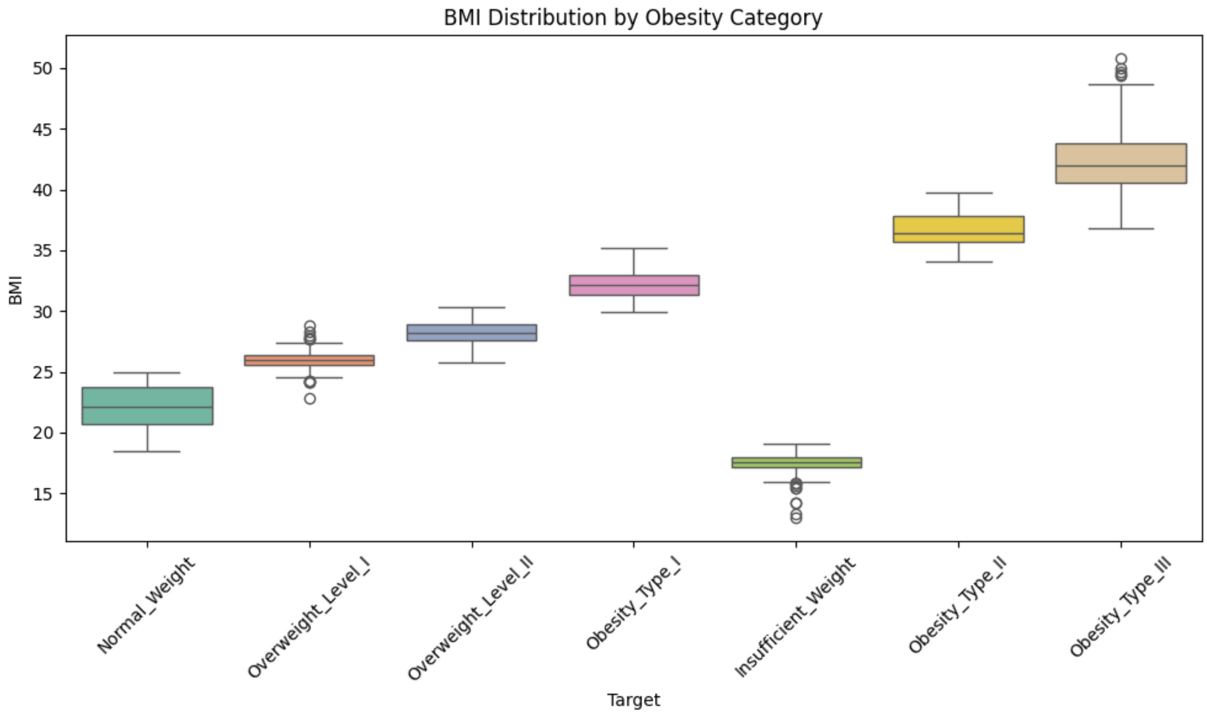


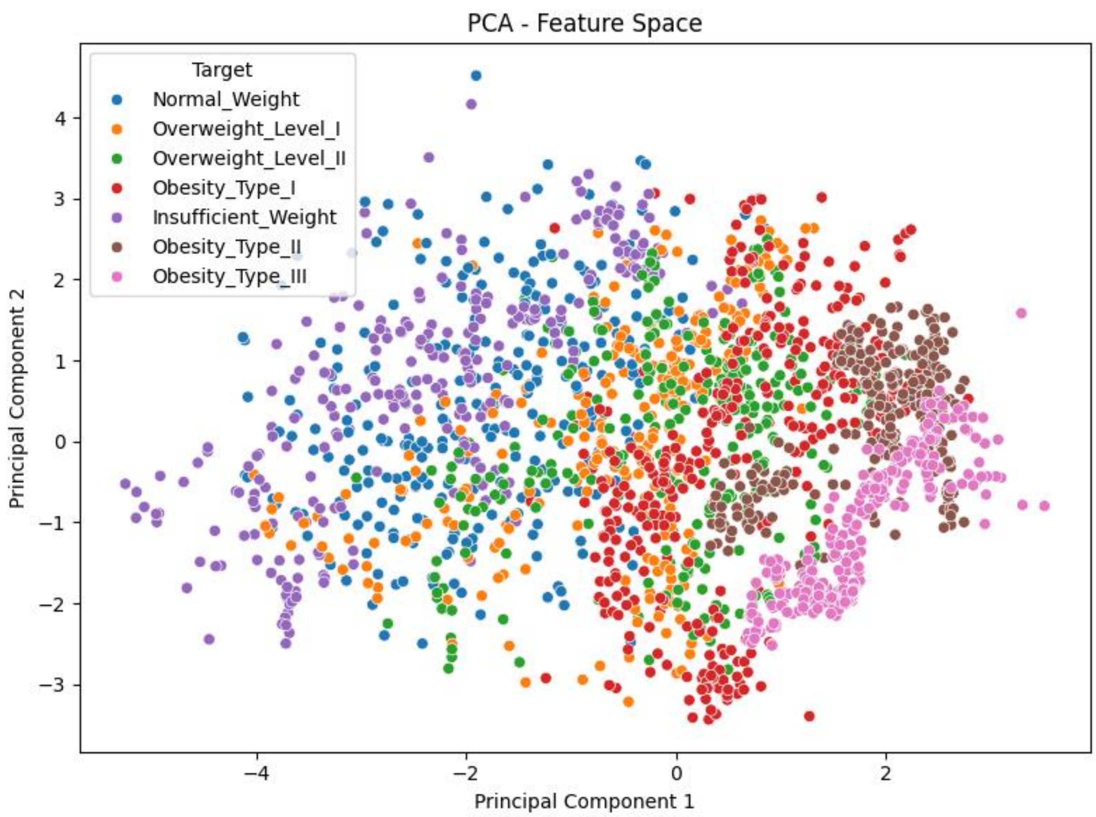

Feature Importance - Tuned Model with BMI
BMI
Gender
Weight
SCC
CAEC
FCVC
CH2O
Height
NCP
SMOKE
family_history
FAVC
CALC
Age
TUE
MTRANS
FAF
Features
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Importance Score

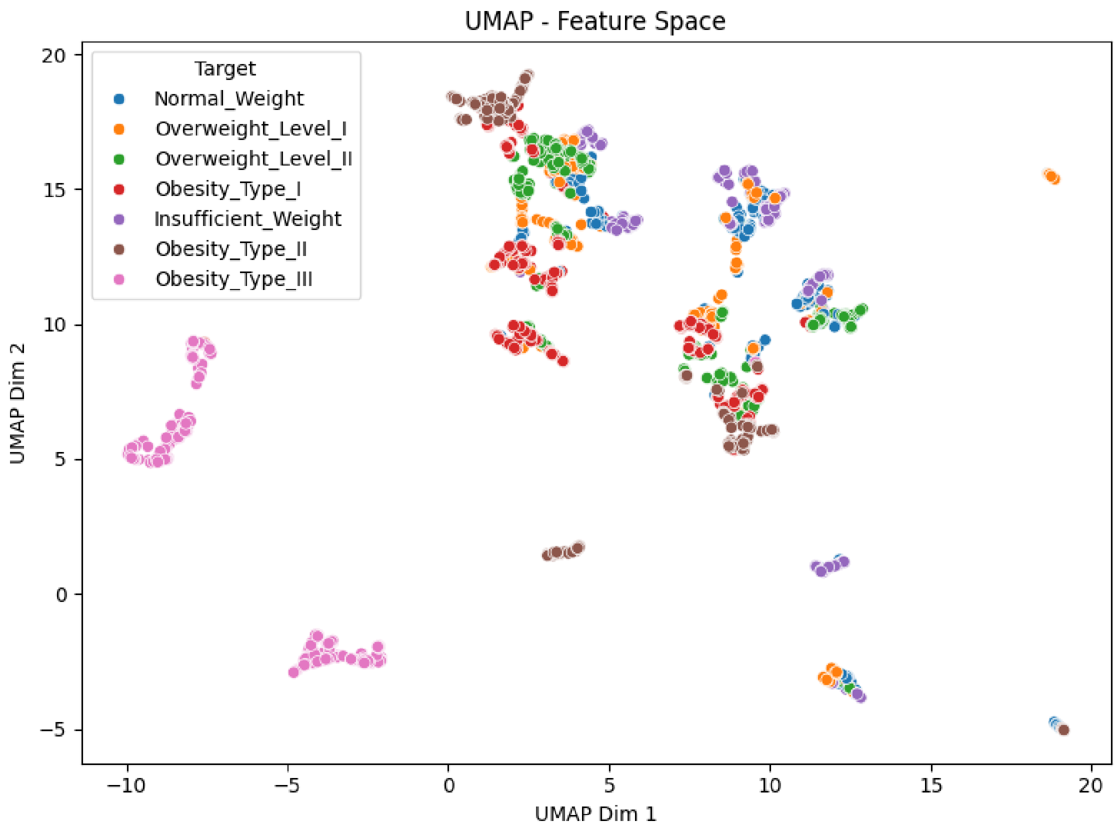

Model Accuracy Comparison
100
98
96
94
92
90
Accuracy (%)
98.58%
95.51%
Basic Model
Tuned Model

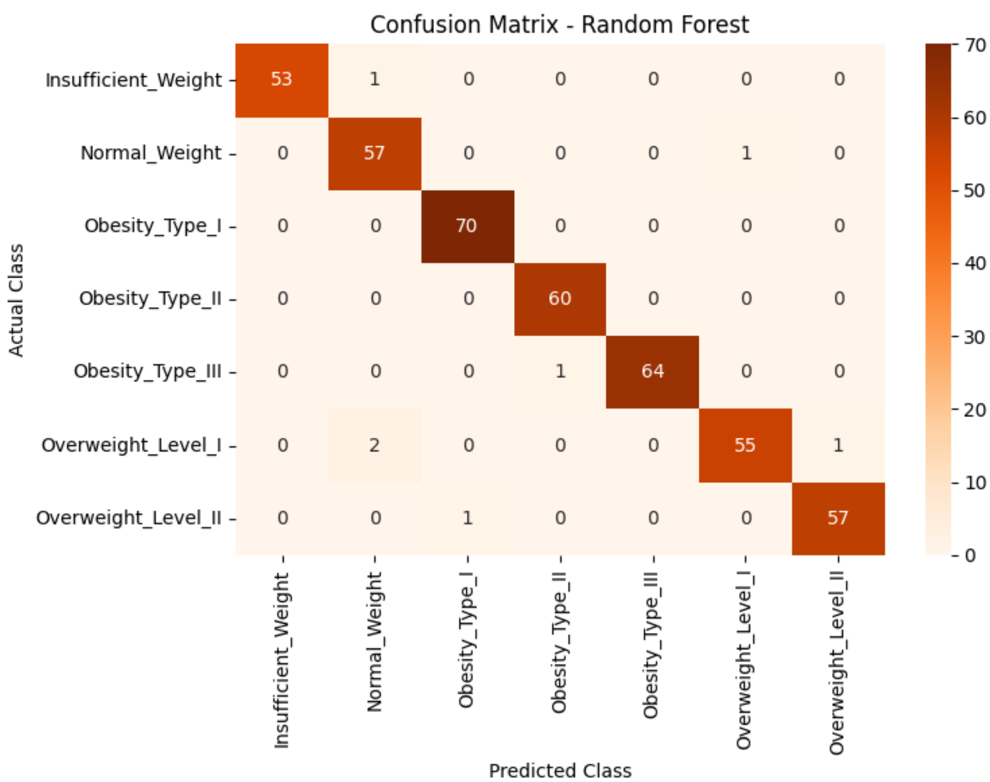

Classification Report Heatmap - Basic Model
precision
recall
f1-score
0.96
0.83
0.97
0.98
1.00
0.96
0.98
0.96
0.96
0.91
0.95
0.99
0.98
0.98
0.90
0.97
0.96
0.95
0.93
0.89
0.98
0.98
0.99
0.93
0.97
0.96
0.95
Insufficient_Weight
Normal_Weight
Obesity_Type_I
Obesity_Type_II
Obesity_Type_III
Overweight_Level_I
Overweight_Level_II
accuracy
macro avg
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
0.84


Classification Report Heatmap - Tuned Model
precision
recall
f1-score
0.98
0.98
1.00
0.98
1.00
0.97
0.98
0.99
0.99
1.00
0.97
1.00
1.00
0.98
0.97
0.98
0.99
0.99
0.99
0.97
1.00
0.99
0.99
0.97
0.98
0.99
0.99
Insufficient_Weight
Normal_Weight
Obesity_Type_I
Obesity_Type_II
Obesity_Type_III
Overweight_Level_I
Overweight_Level_II
accuracy
macro avg
1.000
0.995
0.990
0.985
0.980
0.975
0.970

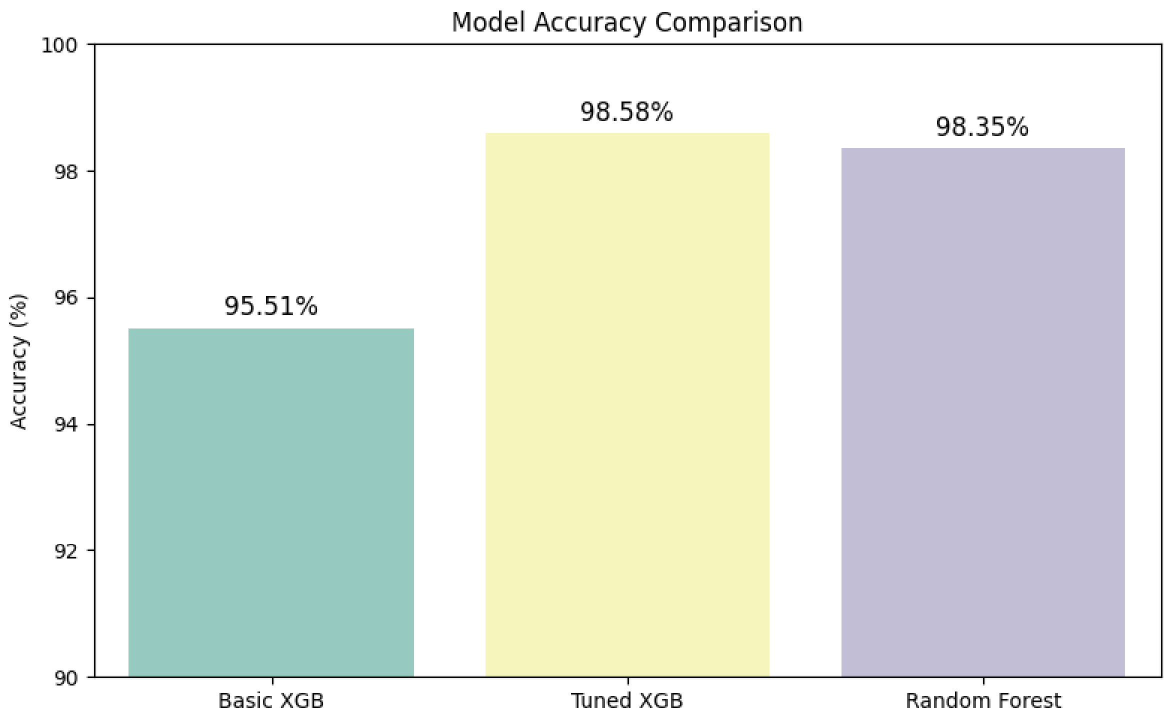

BMI Distribution by Obesity Category
BMI
50
45
40
35
30
25
20
15
Normal_Weight
Overweight_Level_I
Overweight_Level_II
Obesity_Type_I
Insufficient_Weight
Obesity_Type_II
Obesity_Type_III
Target


PCA - Feature Space
Target
Normal_Weight
Overweight_Level_I
Overweight_Level_II
Obesity_Type_I
Insufficient_Weight
Obesity_Type_II
Obesity_Type_III
Principal Component 2
4
3
2
1
0
−1
−2
−3
−4
−2
0
2
Principal Component 1





	Insufficient_Weight	Normal_Weight	Obesity_Type_I	Obesity_Type_II	Obesity_Type_III	Overweight_Level_I	Overweight_Level_II
Insufficient_Weight	53	1	0	0	0	0	0
Normal_Weight	0	57	0	0	0	1	0
Obesity_Type_I	0	0	70	0	0	0	0
Obesity_Type_II	0	0	0	60	0	0	0
Obesity_Type_III	0	0	0	1	64	0	0
Overweight_Level_I	0	2	0	0	0	55	1
Overweight_Level_II	0	0	1	0	0	0	57



Practical Applications

- **Clinical Decision Support**: Assists doctors in identifying at-risk patients.
- **Health Apps**: Powers personalized risk assessments.
- **Public Health**: Informs prevention strategies using visual insights.
- **Research**: Guides studies on lifestyle-obesity relationships.

Challenges and Considerations

- **Synthetic Data**: May introduce noise, requiring real-data validation.
- **Overfitting**: Deeper trees risk overfitting; additional regularization could help.
- **Interpretability**: Visualizations enhance understanding, but SHAP could provide instance-level insights.
- **Scalability**: Grid search is intensive; randomized search could improve efficiency.

Potential Enhancements

1. **Feature Engineering**: Add interaction terms or scale features.
2. **Class Imbalance**: Apply SMOTE if imbalances emerge.
3. **Alternative Algorithms**: Explore LightGBM or neural networks.
4. **Explainability**: Use SHAP for detailed insights.
5. **Data Quality**: Check for outliers or unrealistic values.
6. **Deployment**: Develop an API for real-time predictions, ensuring regulatory compliance.

Conclusion

This project leverages the Obesity Prediction Dataset to build robust models, with the tuned XGBoost achieving 98.58% accuracy. Enhanced visualizations (confusion matrices, feature importance, accuracy comparison, classification report heatmaps, BMI distribution, PCA, UMAP) and Random Forest comparison provide actionable insights into obesity drivers. These support public health strategies and personalized healthcare, with future enhancements improving robustness and deployability.

The Obesity Prediction Dataset Analysis using ML has far-reaching practical applications in clinical decision support, health apps, public health policy, research, personalized healthcare, corporate wellness, and education. Its high-accuracy models (up to 98.58%), robust visualizations (e.g., feature importance, BMI distribution, PCA/UMAP), and comparison with Random Forest provide a reliable foundation for addressing obesity's global challenge. By identifying key predictors like BMI and physical activity, the project informs targeted interventions and personalized recommendations, aligning with its objectives of enhancing public health and contributing to personalized care. Overcoming implementation challenges through data validation, privacy compliance, and user-friendly interfaces will maximize its impact, making it a valuable tool for obesity prevention and management.